

Systematic Evaluation of Dataset Watermarking for Intellectual Protection

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Abstract—This work categorizes existing dataset watermarking approaches based on whether they preserve or disrupt the main task’s predictions. *Prediction-consistent* methods such as Radioactive Data, Data Isotopes, and Style Transformation ensure that predictions of watermarked testing samples still align with the correct task outputs, while *prediction-inconsistent* methods (e.g., typical backdoor-based schemes) intentionally induce misclassifications. Sub-categories are made upon fine-grained criteria, e.g., clean-label, targeted/untargeted, hard-label/confidence verification dependence. Under a unified experimental framework, we revisited all 9 watermarking methods to the year 2025, consisting of five prediction-consistent and four prediction-inconsistent ones. Through end-to-end reproduction and re-implementation, we systematically reassessed their *utility* (i.e., overall accuracy and class-specific accuracy), *verification capability* (i.e., hard-label access and usage of different watermarked samples), and *robustness* under non-adversarial settings (e.g., surrogate model and dataset mixture) and adversarial settings (sample-level and model-level detection). Our study yields several new findings. First, under a unified training-from-scratch evaluation protocol aligned with the original settings of the evaluated methods, most watermarking methods exhibit only minor overall utility degradation (e.g., less than 1%), consistent with their original reports. At the same time, a finer-grained class-wise analysis reveals an overlooked effect: even when the overall accuracy drop is negligible, clean-label watermarking methods can still cause substantial degradation on the target class (e.g., 3.6%). Second, we identify the overlooked data mixture scenario in *non-adversarial* data curation, under which one optimization-based watermarking method and Style Transformation completely fail, revealing a critical but unknown pitfall in a highly practical setting. Third, except for the domain watermark, all the remaining eight watermarking schemes exhibit one or more notable limitations of utility drop, weakened verification performance, and non-robustness.

Index Terms—Dataset watermarking, adversarial attack, robustness, verification.

I. INTRODUCTION

DEEP learning (DL) has made remarkable progress and has become the driving force behind numerous fields, including computer vision [1], large language models [2], autonomous driving [3], and healthcare [4]. The success of DL, however, is not solely attributed to advances in network architectures or training algorithms—it also relies heavily on the availability of large-scale, high-quality datasets [5]. This pivotal role of data has been further amplified by the rise of data-centric AI (DCAI), which advocates for prioritizing data quality over model design [6], [7]. Notably, improving dataset *quality* has been shown to yield greater performance gains for large language models than simply increasing dataset size, which suffers from diminishing returns [8]. In this paradigm, data are not merely fuel for AI, but the determining factor of model capability.

The rising value of data, however, has also intensified concerns over unauthorized use, unconsented redistribution, and misappropriation of datasets [9], [10], [11]. For instance, Clearview AI scraped millions of facial images from social media platforms and services like Venmo to train its facial recognition system [12]. The resulting database, commercialized to law enforcement and private entities, sparked global debates over privacy, surveillance, and ethical misuse. These issues underscore why data protection has long been a critical research domain, reinforced by regulatory frameworks such as the GDPR [13] and the EU AI Act [14].

To mitigate misuse of copyrighted datasets, two main protection strategies have emerged: intrusive dataset watermarking and non-intrusive dataset fingerprinting [9], [11]. Dataset watermarking proactively embeds (imperceptible yet) verifiable signals into the dataset before release. These watermarks function as copyright signatures, intentionally designed to survive typical training procedures and to be reliably triggered during ownership verification. Dataset fingerprinting aims to extract distinctive statistical characteristics or latent distributional patterns from a dataset, treating them as a “natural signature” of ownership. These intrinsic attributes can subsequently be inferred from the behavioral outputs of models trained on the dataset, enabling indirect verification of data provenance without modifying the original data.

Among these, dataset watermarking has attracted particular attention due to its reliability in establishing intellectual property

Received 27 December 2025; revised 25 June 2026; accepted 2 July 2026. Date of publication 7 July 2026; date of current version 1 September 2026. This work was supported by National Natural Science Foundation of China under Grant 62372236 and Grant 62402223. (Corresponding authors: Boyu Kuang; Yansong Gao.)

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Digital Object Identifier 10.1109/TDSC.2026.3710863

TABLE I
SUMMARY OF SATISFACTIONS ACROSS SEVEN KEY DIMENSIONS (TRANSFERABILITY IS ASSOCIATED WITH SURROGATE MODEL)

Watermarking Method	Utility Preservation	Hard-Label Verification	Surrogate Model Requirement	Transferability	Dataset Mixture Robustness	Adversarial Robustness	Data Modality Generalization
T&P [15](2023)	✓	✓	×	N/A	✓	×	✓
T&C [16](2023)	×	✓	✓	✓	✓	×	✓
U&P [17](2022)	✓	✓	×	N/A	✓	×	✓
U&C [17](2022)	×	✓	✓	×	×	✓	×
Radioactive Data [18](2020)	✓	×	✓	✓	✓	✓	✓
Data Isotopes [19](2024)	×	×	×	N/A	✓	✓	✓
Domain Watermark [20](2023)	✓	✓	✓	✓	✓	✓	×
Style Transformation [21](2022)	✓	×	×	N/A	×	✓	×
EntropyMark [22](2025)	✓	×	✓	✓	✓	✓	×

(IP) ownership. In contrast, dataset fingerprinting continues to suffer from reliability challenges, often leading to false or unverifiable claims of ownership [9]. Consequently, dataset watermarking is recognized as a more practical and trustworthy approach for real-world applications [9]. The core idea is to embed subtle, verifiable patterns into the dataset before release. These watermarks act as signatures or identifiers: dormant under normal usage yet reliably detectable through specific means, thereby enabling robust and defensible IP verification. In practical dataset copyright verification scenarios, watermark embedding and verification are both performed by the data contributor, who retains a private set of watermark verification samples. During ownership verification, the contributor queries a suspect model using this retained verification set. The verification goal is to determine whether the contributor’s protected dataset has been used to train the model.

Significant efforts have been devoted to developing dataset watermarking methods [15], [16], [17], [18], [19], [20], [21], [22]. While these methods demonstrate promising performance in terms of utility and verification capability, two critical gaps remain. First, the absence of a unified evaluation framework prevents systematic, end-to-end comparisons, making it difficult to assess their relative strengths and weaknesses or to determine their suitability for different categories of data owners. Second, evaluations have rarely been conducted under practical, real-world settings, which may discourage practitioners from adopting these methods in practice.

Crucially, none of the existing watermarking studies properly account for dataset *curation*, whether the curator is non-adversarial or adversarial. In practice, data curation is the norm: curators aggregate datasets from multiple contributors or sources, and either train models themselves or, more commonly, monetize these datasets by selling them to model providers. This practice underpins the rapidly growing “Data-as-a-Service” (DaaS) paradigm, with established providers such as Appen [23], Clickworker [24], TELUS International [25], and Amazon Mechanical Turk [26].

Considering the central role of curators as the gateway between data contributors and model providers, it is vital that watermarking designs explicitly address curation scenarios. Even in the non-adversarial case, a curator may mix datasets from multiple contributors, leading to what we refer to as the *dataset mixture* problem. No existing method has evaluated robustness under this common setting. As we identified, one

optimization-based dataset watermarking, as well as *Style Transformation* [21], fails in this common and benign setting.

The challenge is even more pressing in the adversarial case, where a curator may apply detection to deliberately remove watermarked samples—motivated by the desire to avoid compensating contributors and thereby maximize profit. When the data curator also trains and publishes models as a service, they can also check whether the model is compromised by applying model-level integrity detection e.g., methods to detect a backdoor. While some existing works [17], [19] considered the adversarial scenarios, their detection methods are outdated. As summarized in Table II in Section III-C, the considered training phase sample-level detection is all in the year of 2018. In contrast, this system study considers the *Beatrix* [27] (NDSS 23’) for such robustness reassessment, as well as *TED* [28] (S&P 24’) and *MM-BD* [29] (S&P 24’). We also note that the latest dataset copyright auditing SoK in IEEE S&P (2025) [9] also neglects this thorough adversarial scenario. This leaves the adversarial scenario largely unexplored, despite being arguably the most relevant to real-world deployment (detailed in Section III-C).

Our Work and Contributions: This work bridges a critical gap between existing dataset watermarking methods (all works to 2025) and their performance under real-world data curation, S&P SoK [9]. We introduce a clearer categorization of watermarking techniques using both coarse-grained criteria (e.g., whether verification relies on prediction consistency with the main task) and fine-grained criteria (e.g., consistency between labels and content of marked samples)—a level of detail absent in [9]. We further discuss conceptual relationships across methods, and reframe the technical underpinnings of prediction-consistent watermarking techniques (e.g., *Radioactive Data* [18], *Data Isotopes* [19], *Domain Watermark* [20], *Style Transformation* [21], and *EntropyMark* [22]) in a more concise and unified way for accessibility (Section III). Importantly, we provide a structured taxonomy of adversarial settings against dataset watermarking and apply all three detective approaches (we do not primarily consider preventive approaches as reasoned in Section III-C) for assessing the watermark robustness. Through a systematic assessment of watermarking, especially under realistic data curation scenarios, our key findings and contributions are as follows:

TABLE II
SUMMARY OF CONSIDERED DATASET WATERMARK DEFENSES

Watermark method	Data cleansing	Robust training	Post-training model detection	Post-training removal	Online input detection and purification
T&P [15]		ABL [44] (2021)		Fine-tune, Model pruning	
T&C [16]					VAE [55] (2015)
U&P, U&C [17]	SS [53] (2018), AC [54] (2018)		NC [47] (2019)	Fine-tune, Model pruning, MCR [56] (2020), NAD [57] (2021)	STRIP [51] (2021), Grad-CAM [58] (2017)
Radioactive data [18]				Distillation	
Data Isotopes [19]	SS [53] (2018), AC [54] (2018)			Fine-tune	Spurious correlation detection [59] (2022)
Domain watermark [20]			NC [47] (2019)	Fine-tune, Model pruning	Scale-UP [60] (2023)
Style transformation [21]				Fine-tune, Model pruning	
EntropyMark [22]	SS [53] (2018)			Model pruning	STRIP [51] (2021)

• *Fine-Grained Accuracy Drop*: We reevaluate the utility impact of dataset watermarking under a unified training-from-scratch evaluation protocol aligned with the original settings of the evaluated methods. Under this setting, most watermarking methods yield only minor overall accuracy drops (e.g., less than 1%), consistent with their original reports. We further uncover a neglected but important effect: while overall accuracy loss is minimal, clean-label watermarking methods can significantly degrade performance on the target class (e.g., 3.6% drop), even when the overall utility drop is negligible (e.g., 0.81%).

• *Non-Adversarial Data Curation*: These approaches, designed to preserve label-content consistency and/or imperceptibility, inherently rely on surrogate models for optimization. We find that their watermarks can weaken when the deployed model diverges from the surrogate architecture. More critically, under dataset mixture, one optimization-based watermark, U&C, as well as Style Transformation, fail completely. Moreover, under class-imbalanced mixture, the verification signals of Radioactive Data and Domain Watermark also degrade notably, further raising concerns about the practical robustness of these methods in realistic non-adversarial data curation settings (Section V-A).

• *Trade-Offs Dilemma*: As summarized in Table I, all watermarking methods except Domain Watermark fail to satisfy at least one property—utility preservation, verification capability (i.e., hard-label, transferability), or resilience to adversarial and non-adversarial curation (data mixture and adversarial scenario). When further considering data modality generalization, it is noted that Domain Watermark is specifically designed for the image modality; it is not directly applicable to other data modalities.

Taken together, these new findings highlight not just weaknesses in specific methods, but a deeper systemic barrier: protecting dataset IP under adversarial data curation remains largely an unresolved and foundational challenge. Our study suggests that meaningful progress will require rethinking watermarking protection strategies and their evaluations when they are designed.

II. TAXONOMY OF RELATED WORK

This section categorizes related work on dataset watermarking for dataset IP protection and discusses their interrelationships. Our evaluations primarily focus on classification tasks. First, classification has been more extensively studied compared to, for example, generative tasks [30]. Second, most existing works on dataset watermarking are also evaluated on classification benchmarks, as noted in [9]. Third, assessing robustness in this relatively mature field provides insights into the effectiveness of dataset watermarking techniques before their deployment in real-world applications.

We do not consider model watermarking for *model IP protection* [31], [32], [33], as dataset watermarking and model watermarking follow distinct threat models [9], [11]. Model watermarking assumes control over model training, which gives the defender (or model owner) stronger protection capabilities—an assumption that is infeasible for dataset watermarking. Consequently, protecting model IP is generally considered easier than protecting dataset IP. For instance, backdoor-based watermarking has been widely used for models, and once the adversary controls training, such backdoors/watermarks can become extremely challenging to detect [34], [35].

A. Dataset Watermarking

A typical dataset watermarking scheme consists of two main steps: dataset watermark injection and watermark verification. Before releasing the dataset, the data owner subtly embeds a watermark into the data, which is designed to be imperceptible during normal use. When ownership verification is required—such as in cases of suspected data misuse—the embedded watermark can be detected using specific techniques.

Categorization Criteria: Depending on whether the test watermark samples used during verification produce predictions that are consistent or inconsistent with the main task of the underlying model, we broadly categorize existing watermarking methods into two groups: *prediction-inconsistent* methods and *prediction-consistent* methods. We further sub-categorize these

methods based on finer criteria, such as whether the watermarked training samples are label-dirty or label-consistent.

It is worth noting that we do not adopt the categorization in [9], which groups watermarking methods as backdoor-based, radioactive data-based, or style transformation-based. This taxonomy does not clearly specify its criteria. For example, it classifies Data Isotopes [19] and Domain Watermark [20] under Radioactive Data [18], even though these approaches are methodologically distinct. In particular, Domain Watermark relies on a fundamentally different concept that is a hard-to-generalize domain or feature region. Nonetheless, we acknowledge that prediction-inconsistent watermarking is often represented by backdoor-based approaches, whether targeted or untargeted.

1) *Prediction-Inconsistent-Based Watermarking*: This category uses a backdoor technique, which was initially proposed as a form of security attack against deep learning, wherein a model is trained to exhibit malicious behavior when exposed to specific trigger-carrying inputs, while maintaining normal performance on clean data [36]. As a flip side, data contributors intentionally inject a small fraction of trigger-carrying samples or poisoned samples into a training dataset for watermarking purposes. Once a model is trained on this watermarked dataset, it will learn to associate the trigger (or marker in the context of watermarking) with specific outputs. The presence of such backdoor behavior in a suspect model can serve as evidence of unauthorized dataset usage.

We further divide these methods using two criteria: (i) whether marked test samples induce a deterministic target label, and (ii) whether marked training samples preserve label-content consistency. This yields four sub-categories: targeted backdoor with poisoned labels (T&P), targeted backdoor with clean labels (T&C), untargeted backdoor with poisoned labels (U&P), and untargeted backdoor with clean labels (U&C).

■ 1) T&P and T&C: T&P stamps triggers on selected samples and changes their labels to a predefined target class. Li et al. [15] adapt typical backdoor triggers, such as BadNets [37] and Blended Attack [38], for dataset watermarking. Verification checks whether marked test samples are consistently classified into the target label. To avoid the visible label-content inconsistency of T&P, T&C [16] keeps labels unchanged and uses optimization to construct clean-label marked samples while preserving the same targeted verification behavior.

■ 2) U&P and U&C: Untargeted backdoor watermarking avoids a fixed target label. Instead, marked test samples induce statistically dispersed or non-deterministic predictions. U&P uses poisoned-label marked samples, whereas U&C keeps labels unchanged and relies on clean-label optimization. Verification follows Li et al. [17], which uses a pairwise *t*-test on prediction dispersivity between marked and unmarked test samples.

2) *Prediction-Consistent Based Watermarking*: For methods in this category, the suspect model's predictions on marked test samples do not conflict with the main task. Specifically, in classification tasks, the hard-label predictions on marked samples remain identical to their ground-truth labels, regardless of the marker's presence.

Within this category, two subtypes can be distinguished based on the verification signal:

- **Hard-label-based verification.** Domain Watermark verifies ownership can be solely through hard-label predictions, making it suitable for black-box verification.

- **Confidence-vector-based verification.** Radioactive Data, Data Isotopes, Style Transformation, and EntropyMark rely on changes in the confidence vector. Verification, therefore, prefers white-box access (particularly for Radioactive Data) or, at a minimum, access to confidence scores to observe the statistical shift introduced by the watermark.

■ 1) In Radioactive Data [18], a small fraction of training images is subtly modified so that their deep feature representations are *shifted* along a *user-specified direction* u_y specific to a target class y . This direction is sampled from a standard Gaussian distribution in the feature space of a fixed, pre-trained network and then normalized. Once it is sampled, it is held by the data contributor as a secret and will not be changed in the rest of the procedures. A marking procedure optimizes pixel-space perturbations that cause the extracted features of marked images to align with the target direction while remaining visually imperceptible. The optimization requires at least a surrogate model. To verify whether a model was trained on such data, the detector projects the features of test images onto the secret class-specific direction u_y . In the white-box setting, marked images exhibit significantly higher projection values than clean ones, enabling statistical detection. In the black-box setting, the shift manifests as consistently lower loss or higher confidence on marked images, allowing detection through output-based comparisons.

■ 2) In a manner akin to Radioactive Data watermarking [18], Data Isotopes [19] embed provenance signals by exploiting salient *spurious features* that affect the confidence of a target class, thereby eliminating the need for in Radioactive Data and pixel-space optimization *surrogate models*. From this perspective, the *random direction* feature u_y [18] can largely be regarded as a spurious feature that the model learns to associate with the targeted class. A simple illustrative example is that “snow” can serve as an easy-to-learn predictive cue for “wolf” when most training images of wolves feature snowy backgrounds [39]. Consequently, if the trained model is queried with a “dog” image from the dog class placed against a snowy background, the hard label will still be “dog” but the confidence score for “wolf” will be significantly higher than for the same “dog” image without snow. This elevated confidence for the target class “wolf” enables effective watermark verification.

■ 3) Domain watermark [20] leverage a hard-to-generalize domain as a marker. The domain, or marker $\tilde{\Delta}$, is a feature region lying outside the feature scope of the protected dataset. In other words, models trained on the dataset will not naturally recognize the feature of $\tilde{\Delta}$. Given images x from the target class y_t , their marked versions x' are optimized to satisfy two objectives. First, in pixel space, x' should be visually indistinguishable from x . Second, in feature space, x' should resemble $\tilde{\Delta}$. Again, the label of x' is consistent with its visual content. This optimization exploits feature collision,

similar to [40]. As a result, a model trained on the watermarked dataset will recognize the hard-to-generalize marker $\tilde{\Delta}$ as y_t , whereas a non-infringing model will often misclassify it.

■ 4) *Style transformation* [21] alters image color in a way that models can learn but people barely notice. Generally, it converts images from RGB to a hue-based color space and applies small, controlled shifts in hue that preserve the original content and labels. If a model is trained on such marked data, these hue patterns leave a detectable trace: when the owner later queries the model with similarly hue-shifted probes, the model's confidence changes in a predictable way, revealing that the protected data was used.

■ 5) *EntropyMark* [22] introduces a prediction-consistent watermark that boosts model confidence on marked samples without altering their final hard-label outputs. It achieves this by iteratively optimizing watermarked samples through gradient matching and adaptive data selection, ensuring that the model subtly internalizes a confidence-shifting pattern while keeping the image content and labels intact. This optimization requires a surrogate model to fine-tune a watermark generator. During the generator fine-tuning an entropy-based constraint is applied to shape a sharper probability distribution on marked samples, creating a consistent reduction in prediction entropy. A suspect model trained on the watermarked data will exhibit a characteristic entropy drop on patched samples, enabling reliable ownership verification in black-box settings.

B. Discussion

As summarized in Table I, five methods—T&C, U&C, Radioactive Data, Domain Watermark, and EntropyMark—require marked sample optimization and therefore depend on knowledge of the victim model during the marking phase. In practice, this implies that at least a surrogate model must be available to enable watermarking. To preserve both content and label consistency, T&C, U&C, Domain Watermark, and EntropyMark all utilize feature collision techniques [40]. However, reliance on surrogate models can make these watermarking methods sensitive to the choice of model architecture used by end model providers. Based on our evaluations in Section V-A, U&C demonstrates weakened marking transferability. In a dataset mixture setting, the surrogate model can only be trained using the contributor's own subset of data; the watermarked samples may not generalize once merged into the curator's full dataset. Our results indicate that U&C, Domain Watermark, and Radioactive Data all experience decreased verification reliability under this scenario, especially U&C, which can trivially fail.

Finally, we note an important distinction in the verification phase. For *Style Transformation*, the exact marked training samples must be reused during verification (see Section IV-D). By contrast, for all other watermarking methods, verification can be performed on test samples different from those originally marked, provided the same marker is present.

■ *Advantages and Limitations of the Proposed Taxonomy:* The proposed taxonomy categorizes dataset watermarking methods based on whether the verification

behavior is consistent with the model's main-task prediction. Its main strength lies in providing a unified behavioral viewpoint for auditing, allowing methods with different technical designs to be compared under a common criterion. In particular, this taxonomy captures whether watermark verification relies on misclassification behavior or on more subtle shifts in model confidence while preserving the predicted label. This taxonomy is most natural for classification tasks, where the output is discriminative and the relationship between prediction and ground-truth label is explicit. This matches the current landscape of dataset watermarking research, where most existing methods are designed and evaluated on image classification benchmarks.

The proposed taxonomy is currently suited to classification tasks, where prediction consistency can be defined with respect to explicit class outputs. For other tasks, especially generative tasks, the same behavioral criterion is less immediate because outputs are often stochastic and may not correspond to a single discriminative label. In such cases, the taxonomy may need to be extended with more suitable notions of consistency, for example, semantic-level consistency, attribute-level preservation, or statistical shifts in output distributions.

III. EXISTING DATASET WATERMARKING AND ADVERSARIAL DETECTION TECHNIQUES

This section outlines the marking and verification phases, intending to help clearly understand their key steps or operations, while avoiding overly complex formulations. This is especially relevant for prediction-consistent watermarking methods [18], [19], [20], which can otherwise be difficult to interpret without a strong background in adversarial examples or data poisoning techniques. Whenever necessary, we use easy-to-understand examples. It also facilitates the analysis of their shortcomings later on.

We then classify the five known adversarial categories that threaten dataset watermarking. We highlight that they were treated superficially in the literature to a large extent when assessing the adversarial robustness of the dataset watermarking.

A. Prediction-Inconsistent Watermarking

Prediction-inconsistent watermarking induces abnormal predictions on marked test samples while preserving normal behavior on clean samples. We summarize their marking and verification procedures below.

■ 1) *T&P* and *T&C*: Both methods are verified by checking whether marked test samples are consistently classified into a predefined target class. They differ in how marked training samples are constructed. *T&P* randomly selects samples, stamps a trigger, and changes their labels to the target class. *T&C* instead marks samples from the target class while keeping their labels unchanged, using optimization to preserve the clean-label property. Thus, both methods can be viewed as dataset-watermark adaptations of targeted backdoor attacks.

It is important to note that both methods are equivalent to typical backdoor attacks. Thus, any data poisoning-based backdoor attack can be adapted to serve the purpose of watermarking.

In other words, data contributors may choose from different trigger types (e.g., patch triggers [37], blend triggers [38]) as well as backdoor variants such as universal backdoors or partial backdoors [41].

■ 2) U&P and U&C: These methods are verified by measuring the dispersibility of predictions on marked test samples: rather than mapping to a fixed target class, marked samples should induce randomized or statistically dispersed predictions. U&P uses poisoned-label marked samples, whereas U&C keeps labels unchanged and optimizes marked samples so that their feature representations deviate from their original class while remaining visually consistent. Similar to T&C, U&C requires a surrogate model for clean-label optimization.

B. Prediction-Consistent Watermarking.

■ 1) Radioactive Data [18] has the marking phase to craft watermarking samples before data release and the verification phase.

Marking phase: A surrogate model that is the pre-trained feature extractor maps each image x to a feature vector $f(x)$. In most cases, the surrogate model is the victim model [18]. For each class y , a random unit vector u_y is sampled from $\mathcal{N}(0, I)$ and normalized. Once u_y is sampled, it will not be changed and serves as a hidden secret. A small scalar α controls the magnitude of the shift. For an image x in class y , a perturbation δ is optimized so that $f(x + \delta) \approx f(x) + \alpha u_y$ under imperceptibility constraints (e.g., small ℓ_2 or ℓ_∞ norm). The α controls the strength, which is set to be small e.g., 0.05. This yields “marked” images whose features are nudged along u_y .

Verification phase: It can be done either in a white-box or a black-box manner. In *white-box*: given a suspect model, extract features for marked and clean samples in class y , compute projections $\langle f'(x), u_y \rangle$, and use a statistical test to check for a significant shift in marked samples. In *black-box*: query the model on marked and clean samples; marked samples typically yield higher confidence or lower loss for the target class, and a gap indicates training on radioactive data.

■ 2) Isotope Data also consists of two phases.

Marking Phase: It selects a target label and designs an isotope marker (a salient, easy-to-learn spurious feature such as a simple patch/texture or a physical object e.g., a black eye-glass in facial recognition). In [19], images from ImageNet are chosen to function as the marker for blending into a small subset of images for a targeted class to create marked images. The blending is controlled through two key parameters of α and p , where α regularizes the transparency (the higher, the more opaque the marker), and p is the marking rate e.g., 1% of the dataset. The label of the marked image remains intact. The trained model learns an association between the mark and the confidence (instead of the hard-label) of the target class during training [19].

Verification Phase (black-box): With only black-box query access, the verifier constructs paired test samples by embedding both the same marker and an external marker into benign (or held-out) images, queries the model, and compares the target label’s confidence between the two. The external marker is drawn from the same distribution as the original marker but

differs from the one used in the marking phase (e.g., a different image from ImageNet), thereby reducing false positives. For models trained on isotope data, the confidence for the target class will be consistently higher when the image carries the same marker. The verifier then aggregates results over all paired samples and applies a hypothesis test; a statistically significant increase in confidence provides evidence that the model was trained on the isotope data.

■ 3) Domain watermark [20] uses $\tilde{\Delta}$ to denote a learned *hard-to-generalize domain* constructed so that models trained only on the protected in-distribution dataset $\mathcal{D}$ perform poorly on the domain $\tilde{\Delta}$, while models trained on the watermarked dataset will generalize to it. The $\tilde{\Delta}$ can be treated as a feature vector in general to ease understanding. The data contributor identifies this domain through optimization and creates a domain generator $G_d(\cdot; \theta)$ such that it produces watermarked data $\tilde{x}$ upon x , where $\tilde{x}$ is under the domain of $\tilde{\Delta}$.

Marking Phase: Given a target class y_t and a feature extractor $\phi(\cdot)$ that is the surrogate model, the data contributor optimizes a subset of class- y_t images x into visually indistinguishable variants x' , aligning their features with the hard domain $\tilde{\Delta}$:

$$x' = \arg \min_{x' \approx x} \|\phi(x') - \tilde{\Delta}\|^2$$

$$\text{s.t. } \|x' - x\| \text{ is small, and } y(x') = y_t, \quad (1)$$

This *feature-collision* style optimization [40] preserves pixel-space fidelity (clean-label, imperceptible) while pulling x' toward $\tilde{\Delta}$ in feature space. The watermarked training set contains these (x', y_t) pairs.

Verification Phase (black-box): With query-only access to a suspect model, the data contributor probes using paired $(x, \tilde{x})$ inputs.¹ A model trained on the watermarked data exhibits consistently higher confidence/accuracy on $(\tilde{x})$ inputs for class y_t than a non-infringing model. Aggregating over probes and applying a hypothesis test yields statistical evidence of ownership.

■ 4) Style Transformation [21] primarily uses *linear color transformation* (LCT) for marking the protected user images.

Marking phase: A user embeds a private signature into a small subset of training images by applying imperceptible yet structured hue rotations in the YIQ color space. The transformation maps each pixel $\mathbf{v}$ via $\mathbf{v}' = \mathbf{g}_k \cdot \mathbf{v}$, where $\mathbf{g}_k = T_{\text{YIQ}} \cdot T_k \cdot T_{\text{YIQ}}^{-1}$ depends on a key k that defines a rotation angle $\Delta_k = \frac{k\pi}{180}$. The crucial factor is the k , where k is selected from a discrete set of angles (e.g., $\{30, 60, \dots, 330\}$). This moves images into a low-density region of the distribution, making them more likely to be memorized by a deep model while remaining visually indistinguishable and label-consistent.

Verification phase (black-box): To determine whether a suspect model f has been trained on watermarked data, the verifier (or arbitrator) attempts to recover the hidden signature key $\hat{k}$ from the candidate space $\mathcal{K}$. Two strategies are considered: (i)

¹In Proposition 1 in [20], the notations of paired inputs are (x, x') , however, the description indeed means $(x, \tilde{x})$ same to workflow illustrated in Fig. 2 of [20]. We thus stick with $(x, \tilde{x})$.

grid search, which exhaustively evaluates all discrete candidates (e.g., $\{30^\circ, 60^\circ, \dots, 330^\circ\}$) and selects the k yielding the lowest prediction loss. This is achieved by applying the k -determined LCT color transformation to the images and feeding them into the suspect model; and (ii) *gradient-based search*, which directly optimizes k through differentiable loss minimization, initialized from multiple random seeds. If the recovered $\hat{k}$ results in significantly lower loss than other candidates, this indicates that the model has memorized the signature and therefore was trained on the protected data. The work of Zou et al. [21] primarily focuses on grid search, since gradient-based search may incur higher computational cost due to the optimization process.

■ 5) EntropyMark [22]

Marking phase: EntropyMark constructs a clean label watermark by introducing a probability-distribution-level bias. Instead of injecting a trigger, EntropyMark fine-tunes a watermark generator using a surrogate model, where the generator is optimized through a gradient-matching objective. Specifically, the generator produced samples are optimized so that the gradient induced during surrogate model fine-tuning aligns with the gradient arising from a patched verification set. In parallel, adaptive data selection is employed to choose samples that are most responsive to optimization, improving watermark injection efficiency. The core mechanism is an entropy-based constraint that explicitly forces the model to output a lower prediction entropy (i.e., a sharper confidence distribution) on marked samples while preserving the original top-1 prediction. As a result, the model learns a stable confidence shifting signature that remains imperceptible to humans and does not disrupt model utility.

Verification Phase (black-box): Given only black-box access, the verifier constructs paired probe samples by attaching the patch to held-out clean images, producing patched and unpatched counterparts for comparison. The verifier then queries the suspect model and compares the prediction entropy between the two sets. Since models trained on EntropyMark data exhibit reduced entropy on marked probes, the verifier performs a statistical hypothesis test on the aggregated entropy differences. A significant entropy drop indicates that the model has learned the EntropyMark signature.

C. Adversarial Scenarios and Defense Option

Dataset watermarking embeds structured patterns into training data to induce verifiable model behaviors, fundamentally acting as a form of data poisoning. Crucially, this applies to both prediction-inconsistent (backdoor-based) and prediction-consistent schemes. While prediction-consistent methods do not enforce hard-label misclassification, they embed subtle features that systematically bias the model's confidence toward a target class. Consequently, they conceptually behave as clean-label backdoors. Anomaly and backdoor detection techniques are suitable for identifying these statistical irregularities during data curation. Based on these defensive strategies, we categorize the existing relevant adversarial scenarios into the following five types.

Data cleansing [27], [42], [43] counters data poisoning attacks or identifies noisy training samples. Since marked training samples can be regarded as poisoned data, data cleansing naturally applies to watermarking.

Robust training [44], [45], [46] aims to train a clean model even when the training set is poisoned. Unlike data cleansing, it does not precisely identify and remove poisoned samples, but rather prevents their effect during training.

Post-training model detection [29], [47] detects whether a trained model is compromised (e.g., backdoored). This setting is common when models are outsourced or obtained from third-party platforms. Detection typically relies on a small held-out set or can even be data-free [29].

Post-training backdoor removal [48], [49] attempts to erase backdoors from a trained model. It is preventive in nature but cannot confirm whether the model is indeed backdoored, and it often reduces model utility.

Online input purification and detection either purifies backdoored inputs to remove malicious behavior at inference [50] or detects adversarial inputs [28], [51], [52]. The former is preventive, while the latter is detective.

The results are summarized in Table II. Existing adversarial settings are often treated superficially by applying naive defenses such as fine-tuning or pruning, both of which are largely ineffective for backdoor removal [15], [17]. In addition, their evaluations are often limited to very early techniques such as Spectral Signatures (SS) [53] and Activation Clustering (AC) [54], both from 2018, which are outdated and trivially fail to cleanse marked samples. While data cleansing is a highly relevant adversarial scenario, the latest SoK [9] does not regard data cleansing as a valid adversarial setting in its evaluations.

In our adversarial evaluation, we adopt three representative detection mechanisms. We do not consider preventive mechanisms because they are applied blindly to any dataset or models. Applying them is often computationally expensive and hurts utility. Detection, then remedy e.g., removal or robust training, is more practical.

Offline sample-level detection: Functioned as an offline poisoned samples detection, Beatrix [27] is developed based on Gram matrices of intermediate-layer activations. Beatrix models high-order feature correlations by computing Gram matrices for samples within each class, and identifies anomalous instances through a kernel-based hypothesis test that measures deviations from the expected class-conditional correlation structure. This mechanism enables Beatrix to detect samples whose internal activation patterns are inconsistent with those of benign data, even when the trigger or manipulation is visually subtle.

Online sample-level detection: We adopt TED (Topological Evolution Dynamics) [28], which detects trigger-carrying (adversarial) inputs by interpreting the neural network as a dynamical system and analyzing the trajectory of an input as it evolves through the model. According to [28], adversarial samples tend to deviate from the manifold of clean inputs and follow a distinct path toward the attacker-specified target region. This topological formulation enables TED to detect adversarial samples during

the online inference phase. Note that it can indeed be performed during training when the model training is completed.

Post-training model-level detection: On top of the above sample-level defenses, we also consider MM-BD [29], a maximum-margin-based post-training backdoor detector that evaluates only the logits of a trained model. MM-BD constructs a maximum-margin statistic from pre-softmax outputs and performs unsupervised anomaly detection to identify classes exhibiting abnormal decision-boundary behavior. Importantly, MM-BD is designed not only to determine whether a model is backdoored, but also to identify the specific target class that the backdoor attempts to steer predictions toward. This enables MM-BD to attribute anomalous behavior to a particular class without relying on any knowledge of the trigger pattern. Furthermore, MM-BD does not require access to clean reference data and assumes no particular structure for the trigger. MM-BD is notably a data-free detection, requiring no reserved clean data.

IV. EVALUATION AND RESULTS

A. Threat Model

Following [9], [18], [19], [20], [21], [22], we assume an adversarial setting in which the data curator acts as the adversary, attempting to detect watermark samples contributed by data owners or whether the trained model has a watermark. By successfully eliminating watermarked samples, the curator can monetize the curated dataset through distribution to model providers without compensating the original contributors, as ownership verification becomes infeasible once the watermark is removed. The curator can also be a model provider, which can detect whether the model contains a watermark before the trained model is released.

Conversely, the data contributor—who publishes the watermarked dataset—is free to adopt any watermarking technique. To ensure stealthiness, however, the watermarking rate must be kept minimal, while still sufficient to enable reliable ownership verification. The effective watermarking rate is especially critical under the dataset mixture scenario (identified later), as watermark strength is naturally attenuated when a curator aggregates data from multiple contributors. Consequently, robust watermarking must retain effectiveness even under low watermarking rates. In addition, the watermarking process must avoid imposing significant degradation on dataset utility, particularly with respect to the predictive accuracy of models trained on the watermarked data.

B. Experimental Setup

Dataset: For comparability and consistency purposes, we primarily use CIFAR-10 as the benchmark dataset for extensive evaluations. This is because all considered watermarking works except [18], [19] have used CIFAR-10 for experimental validations. CIFAR-10 contains 60,000 color images in 10 classes, with 50,000 images for training and 10,000 for testing. Each image size is $32 \times 32 \times 3$. To ensure the generalization, we have also used Tiny-ImageNet, CIFAR-100 datasets for evaluations, as detailed in Section V.

Model Architectures: To ensure consistency with the original implementations, we adopt the model architectures specified in each corresponding watermarking paper. Generally, ResNet-18 [61] is used for all works except [21]—note that [21] uses ResNet-50. Evaluations on other model architectures are in Section V-A. To provide watermarked samples, these watermarking methods [16], [17], [18], [20], [22] (recall summary in Table I) rely on surrogate models for optimization. To evaluate the worst-case scenario from the watermark cleansing perspective, unless otherwise stated, we keep the surrogate model the same as the target model.

Adversarial Detection Configuration: For data-level adversarial detection, both TED and Beatrix require access to a small clean reference set to learn benign feature representation. We reserve 1,000 clean images from CIFAR-10 as the reference samples for both methods. MM-BD operates in a fully post-training setting and does not require access to the training data or any clean reference samples. MM-BD is applied directly to the trained model by analyzing its logits through the maximum-margin statistic.

Evaluation Metrics: To comprehensively assess both the utility and robustness of watermarking techniques, we adopt the following evaluation metrics.

- **Clean Data Accuracy (CDA):** Given a model trained on a watermarked dataset, the CDA measures the model's classification accuracy on benign samples in the absence of the watermark. The CDA of the watermarked model should be on par with that of the clean model trained on a non-watermarked dataset. This metric ensures that watermarking approaches do not compromise the utility of the watermarked dataset.
- **Watermark Verification Capability:** For each watermarking method, we adhere to the verification protocol and success criteria defined in the original work (e.g., watermark success rate, confidence scores, p-values), which may vary across approaches.
- **Adversarial Detect:** For data-level adversarial detection using TED and Beatrix, we evaluate robustness under detection attacks by reporting detection accuracy and false positive rate (FPR). Detection accuracy measures the proportion of watermarked samples correctly identified as anomalous—higher accuracy indicates that the watermark is more easily exposed and removed. FPR reflects the proportion of benign samples incorrectly flagged as anomalous. Therefore, a low FPR is critical for effective watermark cleansing, ideally approaching 0%.

For MM-BD, we evaluate whether the detector can correctly identify the watermark-affected class. For backdoor watermarks, this corresponds to the backdoor target class, whereas for label-consistent watermarking, it corresponds to the class whose samples were marked during watermark injection. We therefore report whether MM-BD identifies the correct target class from the trained model.

C. Prediction-Inconsistent Watermarking

For prediction-inconsistent watermarking, all experiments are conducted on the CIFAR-10 dataset using a ResNet-18 model trained for 120 epochs. The 120 epochs ensure the convergence

TABLE III
ACCURACY DROP (Δ CDA, %) UNDER DIFFERENT WATERMARKING RATES. CLEAN MODEL CDA = 92.43%.

Watermarking Rate (%)	Pred. Incon.					Pred. Con.				
	BadNets	Blended	T&C	U&P	U&C	Radio.	Data Isotopes	Domain Watermark	Style Trans.	EntropyMark
1	-0.40	-0.18	-0.51	-1.34	-2.46	-0.16	-0.44	-0.42	-0.27	-0.36
5	-0.97	-0.66	-1.45	-1.50	-3.16	-0.21	-1.91	-0.68	-0.58	-0.53
10	-1.36	-0.73	-9.18	-2.04	-4.41	-0.45	-10.16	-0.80	-0.81	-0.76

TABLE IV
THE CDA OF T&C ON TARGETED CLASS

Watermarking Rate %	0	1	5	10
Target class CDA	94.20	90.60	84.20	0

TABLE V
WSR OF PREDICTION-INCONSISTENT BASED WATERMARK

Watermarking Rate(%)	Watermark Method				
	BadNets	Blended	T&C	U&P	U&C
1	99.99	92.96	97.46	77.23	13.61
5	100	99.79	98.86	85.18	46.92
10	100	99.99	99.09	87.36	75.90

of the model. Consistent with prior work [15], [17], random horizontal flipping is employed as the default data augmentation strategy during training. Under this setting, the CDA of the model trained without watermarking reaches 92.43%, serving as a reference baseline. That is, the models trained on those marked training datasets should be on par with the CDA baseline to ensure the utility of these watermarking methods. For the watermarking methods, we vary the marking rate (1%, 5%, and 10%) as one of the evaluation factors. From the data contributor perspective, this marking rate stands for budget and stealthiness should be as small as possible.

■ *Utility*: The results are detailed in Table III. As the watermarking rate increases, the model's CDA shows a gradual decline. The clean-label backdoor (T&C and U&C) can result in significant utility drop when the marking rate is high, e.g., 10%. This is exacerbated when the untargeted characteristic is incorporated as in U&C, so that even a small watermarking rate of 1% can render a notable CDA drop of 2.46%.

For T&C, markers are embedded exclusively into samples from the targeted class. To ensure a clean-label setting, all such samples are optimized so that their features from the targeted class (e.g., dog) are disrupted to resemble those of other classes (e.g., cat). Consequently, when the poisoning rate is 10%, all samples of the targeted class are marked (i.e., poisoned), causing the benign features of this class to be severely distorted and leading to a substantial drop in CDA. To further validate this rationale, we evaluate the CDA of the targeted class under T&C, which results are detailed in Table IV. As expected, at a 10% marking rate, the targeted class features are completely destroyed, resulting in a 0% CDA. Even at a 1% marking rate, the targeted class suffers a 3.6% drop in CDA, despite the overall CDA drop being only 0.51%, which may appear negligible. This demonstrates that the utility degradation for the targeted class is essentially non-negligible.

For U&P, the marked samples are essentially equivalent to noisy samples since their labels are randomly assigned. This can relatively harm utility due to the noisy-label effect. However, deep learning models are generally robust to a relatively large fraction of noisy labels [62], owing to their strong capacity

for feature generalization. As a result, the utility drop remains insignificant even as the marking rate increases.

For U&C, the utility drop is more pronounced compared to U&P: In addition to the noisy-label effect, U&C also suffers from the same feature-disruption issue observed in T&C. Nonetheless, since the marked samples can be randomly drawn from any class—rather than exclusively from a single class as in T&C—the overall degradation is less severe. Specifically, at a 10% marking rate, the utility drop is 4.71%, which is considerably smaller than that of T&C, as the benign features of each class in U&C are still largely retained. More specifically, on average, in U&C, 1% samples from each class are marked and their class-tied features are destroyed.

■ *Watermark Verification Capability*: The watermark verification capability is directly related to the WSR for prediction-inconsistent watermark. The definition of WSR varies depending on the specific watermarking method:

- For T&P and T&C, WSR is the proportion of marked testing samples that are classified into the target class.
- For U&P and U&C, WSR is defined as the proportion of testing samples that are correctly classified in their clean form, but are misclassified once the trigger is added.

The WSR results are detailed in Table V. For BadNets, and Blended belong to T&P, and T&C, a small marking rate of 1% achieves a WSR easily exceeding 90%. U&P and U&C requires higher marking rates to reach a comparable WSR. For instance, the WSR is 77.23% for U&P and only 13.61% for U&C.

Takeaway 1: By considering the utility and WSR, the T&P and U&P methods appear to stand out to maintain a comparable CDA while providing strong verification capability. However, both cannot retain the desirable clean-label marked training sample

■ *Adversarial Robustness*: We evaluate the adversarial robustness of prediction-inconsistent watermarking

TABLE VI

DETECTION PERFORMANCE OF PREDICTION-INCONSISTENT WATERMARKING METHODS UNDER ADVERSARIAL SCENARIOS

Detector	Preset FPR(%)	Results	Watermark Method				
			BadNet	Blended	T&C	U&P	U&C
TED	1	Det. Acc(%)	94.92	92.52	94.20	87.56	6.96
		Online FPR(%)	0.36	1.56	0.32	0.84	1.32
	3	Det. Acc(%)	98.64	95.40	96.24	88.92	12.68
		Online FPR(%)	1.16	3.04	1.12	2.04	2.64
	5	Det. Acc(%)	99.24	96.44	97.32	89.56	17.56
		Online FPR(%)	2.40	5.20	1.92	3.88	5.12
Beatrix	1	Det. Acc(%)	61.24	58.20	1.28	14.16	3.72
		Online FPR(%)	0.36	0.72	0.60	0.58	0.80
	3	Det. Acc(%)	88.32	82.64	20.68	26.00	10.08
		Online FPR(%)	1.84	2.24	2.12	1.86	2.35
	5	Det. Acc(%)	93.40	93.28	26.00	35.92	13.76
		Online FPR(%)	3.23	3.52	3.45	3.73	3.94
MM-BD	—	Det. Success	✓	✓	✓	×	×

methods under TED, Beatrix, and MM-BD, respectively. Table VI summarizes the detection performance. For the data-level detectors (TED and Beatrix), both detectors require specifying a preset FPR during calibration, which determines the operating threshold used for distinguishing anomalous samples. During evaluation, we measure the detection accuracy and online FPR, i.e., the actual proportion of benign samples mistakenly flagged when the detector is applied to the test data. We evaluate both detectors under preset FPR levels of 1%, 3%, and 5%, and report results on datasets watermarked at a 5% marking rate. For model-level inspection, MM-BD evaluates the trained classifier and determines which class displays abnormal decision-boundary behavior. We treat detection as successful only when MM-BD correctly identifies the watermark-affected class.

From Table VI, we observe that TED consistently achieves high detection accuracy on most watermark types, except for U&C. This result aligns with the low watermark success rate (WSR) of U&C: with only a small fraction of samples effectively marked.

Beatrix exhibits a more selective detection pattern. It performs robustly on BadNet and Blended, both of which belong to the classical targeted and poisoned label backdoor and typically induce stronger anomalies in feature-space. In contrast, Beatrix yields limited detection accuracy on T&C, U&P, and U&C. Despite the low accuracy, the non-negligible anomaly samples still suggest the presence of distributional irregularities, offering a subtle indicator that the dataset contains manipulated samples—even if the detector cannot reliably isolate them.

For model-level inspection, MM-BD successfully identifies the watermark-affected target class for BadNet, Blended, and T&C, confirming that these watermarking schemes leave detectable decision-boundary distortions.

Overall, these results demonstrate that prediction-inconsistent watermarks can be partially detected through modern data-level and model-level defenses. Strong targeted watermarks tend to leave clear signatures detectable by both TED and MM-BD,

TABLE VII

TAGGED-CLASS ACCURACY AND WATERMARK SUCCESS RATE OF Data Isotopes UNDER DIFFERENT WATERMARK INJECTION STRENGTHS α AND WATERMARKING RATES; THE CLEAN TAGGED-CLASS ACCURACY IS 92.50%

Watermarking Rate	$\alpha = 0.2$		$\alpha = 0.4$		$\alpha = 0.6$	
	TCA(%)	WSR(%)	TCA(%)	WSR(%)	TCA(%)	WSR(%)
1%	91.20	20	89.70	100	88.50	100
5%	87.60	100	83.60	100	82.30	100
10%	0	100	0	100	0	100

whereas utility-preserving or weak-marking schemes create only subtle or inconsistent anomalies, making them significantly harder to detect reliably.

Takeaway 2: Prediction-inconsistent watermarks vary in adversarial robustness: strong targeted schemes leave detectable artifacts, while weak or utility-preserving schemes (i.e., U&C) introduce only subtle deviations that current detectors struggle to capture. Unfortunately, by considering utility, the most stealthier U&C harms the utility most

D. Prediction-Consistent Watermarking

All models are trained for 120 epochs on ResNet-18 following the setting in Section IV-C.

■ *Utility:* Table III further reports the CDA of each watermarking method under different watermarking rates. All methods, except Data Isotopes exhibit negligible utility degradation (no more than 0.81%), even at a 10% marking rate. In contrast, Data Isotopes shows a pronounced CDA drop as the marking rate increases. This is because it embeds markers into samples from a single class and leverages optimization to satisfy the clean-label requirement. When all samples from the chosen/targeted class are marked, the benign features of that class are severely disrupted, leading to significant degradation. The underlying reason is the same as for T&C, as detailed in Section IV-C.

To further illustrate this class-specific utility effect, Table VII reports the tagged-class accuracy (TCA) and watermark success rate (WSR) of Data Isotopes under different watermark injection strengths α and watermarking rates. Here, α controls the visibility of the isotope mark by adjusting its blending weight in the marked image [19]. The results show that the utility impact of Data Isotopes is mainly concentrated on the tagged class: as α or the watermarking rate increases, the tagged-class accuracy drops markedly while watermark verification becomes more effective.

■ *Watermark Verification Capability:* Each of the methods utilizes a different metric for quantifying the verification capability.

Radioactive watermark: Its verification process is conducted via statistical means as a binary hypothesis test, which is formulated as:

• *Null hypothesis $\mathcal{H}_0$:* The suspicious model $\mathcal{M}$ is trained on clean data.

TABLE VIII
STATISTICAL VERIFICATION CAPABILITY OF RADIOACTIVE WATERMARK [18]
AS A RELATIONSHIP WITH WATERMARKING RATE: HIGHER $-\log(p)$,
STRONGER CAPABILITY

Watermarking Rate(%)	0	1	5	10
$-\log(p) \uparrow$	0.47	1.33	1.51	2.68

- *Alternative hypothesis $\mathcal{H}_1$* : The model $\mathcal{M}$ is trained on marked data.

The verification computes a p -value, indicating the probability of observing the model's behavior under $\mathcal{H}_0$. A smaller p -value (i.e., a larger $-\log(p)$ value) indicates higher confidence that the model was trained on radioactive data. Thus, $-\log(p)$ is a quantitative watermark verification score—a higher score suggests stronger evidence of unauthorized dataset usage. Following the setting in [18], we perform the verification using vanilla images from the held-out validation set in CIFAR-10.

Table VIII presents the $-\log(p)$ values of Radioactive Data under different watermarking rates. A higher $-\log(p)$ value, resulting from a higher watermarking rate, indicates stronger statistical evidence that the model was trained on marked data.

Data isotopes: Its verification is built upon paired one-sided Student's t -tests, focusing on measuring distributional shifts in the model's output confidence. Specifically, given a set of auxiliary testing images, two probes of each image are created by applying two different markers: t and t' . The former is the one used during marking training data, while the latter is an external marker. The test compares the distributions of the predicted probabilities of the target class for the two sets. The verification is also based on a binary hypothesis.

- *Null hypothesis $\mathcal{H}_0$* : The mean/average of the predicted confidence scores for marks t and t' are equal, implying the model $\mathcal{M}$ was trained on a clean dataset.

- *Alternative hypothesis $\mathcal{H}_1$* : The mean for mark t is significantly higher than that for t' , implying the model $\mathcal{M}$ was trained on images containing mark t .

We follow the verification parameters from [19] using the default watermark injection strength $\alpha = 0.4$, where verification is performed over 5 rounds of t -tests on 250 test images sampled from the auxiliary dataset—this means the number of model queries is $5 \times 250 \times 2 = 2500$ with $\times 2$ accounting for the paired probes per image. If the p -value from the t -test remains below the threshold across multiple rounds, the verifier concludes that the watermark exists in the model's training data. At a watermarking rate of 1%, Data Isotopes achieve perfect verification performance, with a true positive rate of 100% (correctly identifying infringed models) and a false positive rate of 0% (never misidentifying a clean model as infringed). This demonstrates that Data Isotopes can deliver reliable watermark verification even when only a small fraction of the dataset is watermarked.

Domain watermark: Its verification is directly characterized by the WSR. The WSR of Domain Watermark is the proportion of marked data generated by the domain generator being classified into their ground-truth label (recall

TABLE IX
WSR of Domain Watermark UNDER VARYING WATERMARKING RATE

Watermarking Rate (%)	1	5	10
WSR (%)	98.10	98.90	99.60

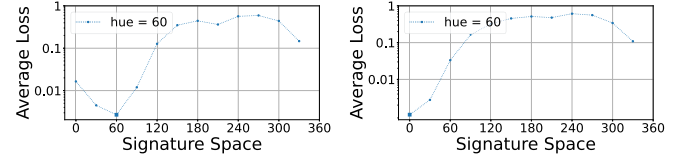


Fig. 1. Marker/signature resulted loss curve for Style Transformation. (Left) By using grid-search, the minimum occurs at 60° , matching the true signature used for watermark embedding. (Right) Signature verification using randomly selected images.

Section III-B). As the marked data is from the hard-to-generalize feature region, only the model $\mathcal{M}$ trained on the protected dataset can exhibit a high WSR; otherwise, a very low WSR.

The WSR of Domain Watermark under different watermarking rates is detailed in Table IX. The WSR easily reaches 98.10% even given a 1% watermarking rate. This means that Domain Watermark can indeed provide strong verification capability with a low watermarking budget.

Style transformation: Generally, this method treats the hue rotation degree as its key, signature, or marker. Following their paper [21], we set 60° as the marker, resulting in a key space of 12—this key size is essentially small. Verification can be performed either via grid search or gradient-based search (recall Section III-B). As their paper primarily adopts the former, we also follow the grid search approach. The key recovery results are shown in Fig. 1 (left) under a 1% watermarking rate. When the searched key is 60° , the loss is significantly lower, indicating that the model contains the watermark and the dataset ownership can be reliably claimed. We also applied gradient-based search. However, we found that it fails to correctly recover the key: the identified key is at 28° , which lies outside the expected ground-truth range of $(45^\circ, 75^\circ)$.

We note that Style Transformation imposes a stringent requirement on the images used for verification: they must be drawn from the exact set of images used in the marking phase. Otherwise, the signature cannot be correctly identified. For instance, under the same setting but with different images, the recovered key is 0° as shown in Fig. 1 (right), which deviates from the ground truth.

This requirement is clearly *problematic* in practice. Consider the case where the data curator only adopts 50% of the contributor's dataset and mixes it with data from other contributors—a dataset mixture scenario discussed further in Section V-A. In this case, only 50% of the marked images would actually contribute to training the model. However, during verification, all marked images would be used, substantially undermining verification reliability, since half of them were never seen by the model.

EntropyMark: It verifies ownership by measuring the entropy shift between benign samples x and their patched versions

TABLE X
P-VALUES OF THE PAIRED t -TEST FOR EntropyMark UNDER DIFFERENT
WATERMARK RATES

Watermark Rate(%)	0	1	5	10
P-value	0.9998	0.0014	0.0002	0.0000

x' . Let $E_b = H(F(x))$ and $E_w = H(F(x'))$ denote the prediction entropies of a suspicious model F . Here, the entropy $H(\cdot)$ is computed as

$$H(x) = - \sum_{i=1}^N p_i \log p_i, \quad (2)$$

where p_i is the predicted probability that x belongs to the i -th class, and N is the number of classes.

Ownership is tested via the hypothesis

$$\mathcal{H}_0 : E_b + \tau = E_w \quad \text{vs.} \quad \mathcal{H}_1 : E_b + \tau > E_w, \quad (3)$$

where $\tau \in [-1, 1]$ controls the decision margin. In practice, a set of benign–patched sample pairs is evaluated using a paired t -test, and the resulting p -value determines whether $\mathcal{H}_0$ is rejected.

In practice, a set of benign–patched pairs is used to conduct a paired t -test, and the resulting p -value determines whether $\mathcal{H}_0$ is rejected. As summarized in Table X, when no watermark is present, the p -value remains close to 1 (0.9998). At a low watermark rate of 1%, the p -value already drops to 0.0014, clearly separating it from the clean model. As the watermark becomes more prominent (5% and 10%), the p -value further decreases to 0.0002 and 0.0000, respectively, strongly rejecting $\mathcal{H}_0$ and confirming that the suspicious model was trained on the protected dataset.

Takeaway 3: By considering the utility and WSR, the Do-main Watermark and EntropyMark methods appear to stand out to maintain a comparable CDA while providing strong verification capability

■ Adversarial Robustness:

Prediction-consistent watermarking schemes exhibit strong robustness against both data-level and model-level adversarial detection. As shown in Table XI, the detection accuracy of TED remains extremely low across all preset FPR levels. Among the five schemes, EntropyMark shows the highest detectability under TED, but the accuracy remains modest (about 12% at 5% preset FPR), confirming that its entropy-shift signal is still far less detectable compared to traditional backdoor triggers. For Beatrix, across all preset FPR settings, none of the methods exceeds 8% detection accuracy, and most remain close to random guessing. This suggests that prediction-consistent watermarks do not disrupt feature covariance structures in a way that Beatrix can systematically exploit.

At the model level, MM-BD fails to identify the watermark-affected class for all five methods. This result highlights a fundamental robustness advantage of prediction-consistent watermarking: because these schemes do not induce evident class decision-boundary shifts, they do not trigger MM-BD's

TABLE XI
DETECTION PERFORMANCE OF PREDICTION-INCONSISTENT WATERMARKING
METHODS UNDER ADVERSARIAL SCENARIOS

Detector	Preset FPR(%)	Results	Watermarked Method				
			Radio.	Data Isotopes	Style Trans.	Domain	Entropy Mark
TED	1	Det. Acc(%)	1.84	1.28	1.12	1.32	2.48
		Online FPR(%)	0.76	1.16	0.72	1.36	1.56
	3	Det. Acc(%)	3.24	3.52	2.72	3.00	6.16
		Online FPR(%)	1.48	3.16	2.44	2.72	3.24
	5	Det. Acc(%)	7.12	6.20	4.60	8.64	12.60
		Online FPR(%)	4.16	5.32	4.84	4.60	5.36
Beatrix	1	Det. Acc(%)	0.88	0.60	0.68	0.24	1.12
		Online FPR(%)	0.50	0.81	0.72	0.73	1.23
	3	Det. Acc(%)	3.68	2.64	1.76	0.88	3.16
		Online FPR(%)	2.60	2.10	2.03	2.14	2.62
	5	Det. Acc(%)	7.96	4.12	2.52	1.36	5.56
		Online FPR(%)	3.77	3.70	2.92	3.20	4.34
MM-BD	–	Det. Success	×	×	×	×	×

maximum-margin anomaly signal. Among the evaluated methods, EntropyMark is the most detectable, but its signals remain weak relative to standard poisoning patterns, indicating that its improved verification capability does not come at the cost of significantly reduced adversarial robustness.

Takeaway 4: Prediction-consistent watermarks exhibit strong adversarial robustness: none of the evaluated detectors can reliably identify them

V. DISCUSSION

A. Non-Adversarial Robustness Evaluation

In real-world model training pipelines, the data curator may (1) use a model architecture that differs from the one assumed by the data contributor or (2) mix data from multiple sources, leading to partial visibility or distribution mismatches. These factors can inadvertently weaken watermark signals even without deliberate removal attempts. Therefore, we assess the non-adversarial robustness of watermarking methods that rely on surrogate models, focusing on two key dimensions: *model-architecture transferability* and *dataset mixture/reduction*.

We evaluate five watermarking schemes that depend on surrogate-model-based optimization: T&C, U&C, Radioactive Data, Domain Watermark, and EntropyMark. For Style Transformation, which does not rely on surrogate-model optimization, we conduct a reduced-data experiment to examine its sensitivity to the exact set of marked verification samples.

1) *Model-Architecture Transferability:* To evaluate whether watermark signals persist when the curator trains a model with a different architecture, we generate marked samples using a ResNet-18 surrogate model (5% watermarking rate) and then train a VGG-16 model on the resulting watermarked dataset. As shown in Table XII, most watermarking methods maintain strong task utility (CDA) and verification performance (WSR). However, U&C exhibits a substantial WSR drop (50.90%), confirming

TABLE XII
EVALUATION OF WATERMARK TRANSFERABILITY UNDER DIFFERENT MODEL ARCHITECTURES

Metric	Watermark Method				
	T&C	U&C	Radioactive	Domain	EntropyMark
CDA (%)	91.17	88.87	92.82	90.85	91.22
WSR (%)	96.91	50.90	$-\log(p) = 1.85$	99.30	$p = 0.0053$

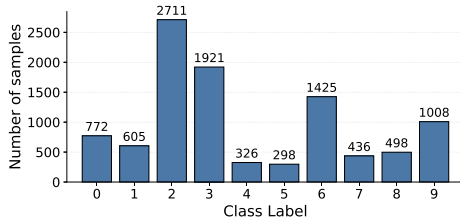


Fig. 2. Class distribution of the watermark-bearing contributor under Dirichlet partition ($\alpha = 2$) on CIFAR-10.

that its optimization is closely tied to the surrogate model's feature geometry and transfers poorly across architectures.

2) *Dataset Mixture and Dataset Reduction*: We further evaluate robustness under dataset mixture, a common scenario where the curator aggregates data from multiple contributors. Since the surrogate model is trained using only the data protector's subset, the optimized marked samples may not generalize when mixed with unseen data distributions. We evaluate three mixture settings using CIFAR-10: (a) the protector owns half of the samples from every class, (b) the protector owns all samples from 5 classes only (CIFAR-10 has 10 classes), or (c) the protector owns an imbalanced subset across classes. For case (c), we consider a more realistic setting where the watermark-bearing contributor may not have a class-balanced dataset due to collection bias or deployment-specific sampling. Specifically, we fix the protector subset size at 10,000 samples and construct class-imbalanced subsets using a Dirichlet distribution [63] with concentration parameter $\alpha = 2$. The resulting class distribution of the watermark-bearing contributor is illustrated in Fig. 2. In all settings, watermarked samples are generated only from the protector subset, while the final model is trained on the full aggregated dataset. To ensure fair comparison, we keep the overall watermarking rate fixed with respect to the full training set at 5% and 1%.

Results are reported in Table XIII. U&C suffers severe degradation across all settings. For prediction-consistent approaches, Radioactive exhibits much weaker statistical signals under Case (c), as reflected by near-zero $-\log(p)$ values. Domain Watermark also shows a noticeable reduction in WSR under class imbalance, especially at lower watermarking rates, while T&C maintains relatively stable verification performance across the three settings.

To better understand the degradation of watermark effectiveness under surrogate-model mismatch and dataset mixture, we analyze the transferability of watermark signals in these scenarios. Specifically, we analyze the internal decision discrepancies between the surrogate model used for watermark optimization

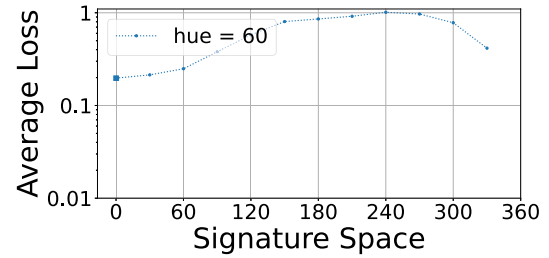


Fig. 3. Watermark verification result of Style Transformation under the dataset mixture scenario where only 50% of the dataset is adopted for training. The verification is conducted using the full set of watermark samples owned by the data contributor.

and the downstream model on the same set of watermark samples. We use the final-layer outputs (logits) to characterize these differences [64], as they reflect the model's internal confidence structure across classes rather than only the final predicted labels. In this sense, the consistency of logits directly reflects whether the watermark-induced response patterns learned by the surrogate model are preserved after transfer. We consider U&C and Domain Watermark under the same settings used above. For surrogate-model mismatch, watermark samples are optimized using a ResNet-18 surrogate model, while the downstream model is trained with a different architecture (VGG-16). For the dataset mixture, the surrogate model is trained on the imbalanced protector subset in Case (c), whereas the downstream model is trained on the full aggregated dataset. As shown in Table XIV, although the two models produce highly consistent top-1 predictions on watermark samples, their logits still exhibit noticeable discrepancies. This indicates that the downstream model does not preserve the surrogate model's watermark-specific logits patterns in a consistent manner. Since these watermark samples are explicitly optimized to induce specific logits behavior in the surrogate model, their effectiveness depends on whether the same behavior can transfer to the downstream model.

For the Style Transformation, we simulate the case where the data curator only uses 50% of the contributor's dataset for training. In this setting, only half of the watermark samples are actually included in the training process, while the data owner still uses the same watermark samples for verification. In this setting, the recovered signature resides at 0° as shown in Fig. 3—ground-truth is 60° , fails for verification. Because a significant portion of the watermark samples were never seen by the model during training. This highlights the limitation of Style Transformation that requires exact same marking images for watermark verification.

Takeaway 5: Watermarking methods—particularly U&C, Radioactive Data, Domain Watermark and Style Transformation—are shown to be non-robust under non-adversarial data curation scenarios, such as changes in model architecture or dataset mixture

TABLE XIII
EVALUATION OF WATERMARK TRANSFERABILITY UNDER DATASET MIXTURES

Watermark Method	Watermarking Rate	Case (a): Partial but Complete Classes		Case (b): Incomplete Classes		Case (c): Imbalanced Classes	
		CDA (%)	WSR (%)	CDA (%)	WSR (%)	CDA (%)	WSR (%)
T&C	1%	91.73	96.63	91.64	95.88	91.92	93.36
	5%	91.44	98.85	91.73	98.26	91.37	97.41
U&C	1%	89.45	8.41	89.17	12.54	89.68	7.22
	5%	87.03	10.39	88.73	15.17	87.24	9.34
Radioactive	1%	92.12	$-\log(p) = 1.04$	92.03	$-\log(p) = 1.28$	91.83	$-\log(p) = 0$
	5%	91.82	$-\log(p) = 1.30$	91.71	$-\log(p) = 1.42$	91.34	$-\log(p) = 0.03$
Domain	1%	91.68	78.90	91.86	92.50	91.92	34.50
	5%	91.21	93.40	91.37	95.80	91.84	54.30
EntropyMark	1%	91.86	$p = 0.0022$	91.92	$p = 0.0153$	92.02	$p = 0.0577$
	5%	91.57	$p = 0.0000$	91.43	$p = 0.0014$	91.68	$p = 0.0000$

TABLE XIV
LOGIT MISMATCH UNDER ARCHITECTURE MISMATCH AND DATASET MIXTURE

Scenario	Watermark Method	Top-1 Consistency(%) $\uparrow$	Logits Similarity(%) $\uparrow$
Arch. Mismatch	U&C	99.0	86.8
	Domain	100.0	88.2
Dataset Mixture	U&C	99.2	84.6
	Domain	100.0	83.5

TABLE XV
EVALUATION OF WATERMARKING METHODS ON TINY-IMAGENET AND CIFAR-100

Watermark Method	Tiny-ImageNet		CIFAR-100	
	CDA(%)	WSR(%)	CDA(%)	WSR(%)
BadNets (T&P)	52.79	100	74.21	99.95
T&C	52.17	85.28	73.64	92.31
U&P	53.11	73.22	74.53	78.68
U&C	51.86	22.45	73.22	15.03
Radioactive	52.92	$-\log(p) = 0.24$	73.87	$-\log(p) = 0.06$
Data Isotopes	51.55	100	74.09	100
Domain	52.62	48.00	73.41	56.00
Style Trans.	53.42	—	74.46	—
EntropyMark	52.43	$p = 0.3276$	74.28	$p = 0.5680$

B. Complicated Dataset

In addition to CIFAR-10, which has been extensively evaluated, Tiny-ImageNet [65] and CIFAR-100 [66] are also widely used benchmark datasets for dataset watermarking, as noted in [9]. Tiny-ImageNet is a subset of ImageNet with 200 classes, containing 100,000 training images and 10,000 test images. In our experiments, the input size to the model is $96 \times 96 \times 3$. CIFAR-100 consists of 100 classes, with 50,000 training images and 10,000 test images, each of size $32 \times 32 \times 3$.

We extend our evaluation to these two datasets. For both Tiny-ImageNet and CIFAR-100, a ResNet-18 model is trained for 150 epochs. The marking rate is kept low at 1%. For T&C and Data Isotopes, however, the maximum marking rate on Tiny-ImageNet is 0.5% ($\frac{1}{200}$), which we adopt, since it embeds markers exclusively into samples from a single target class, and there are 200 classes in Tiny-ImageNet.

The results on Tiny-ImageNet and CIFAR-100 are summarized in Table XV. Backdoor-based methods such as T&P and

T&C maintain strong verification performance, achieving high WSR with limited impact on clean-data accuracy. In contrast, optimization-based methods show weaker robustness, particularly U&C, whose WSR drops significantly on both datasets. Prediction-consistent approaches, including Radioactive and EntropyMark, exhibit weak verification signals, as indicated by the small $-\log(p)$ values and large p -values. Domain Watermark also shows reduced verification performance. For Style Transformation, WSR is not reported since its verification protocol relies on reusing the exact marked samples.

C. Watermark Modality Generalization

Most existing dataset watermarking methods are evaluated primarily in the image domain, and only a few works explicitly consider other modalities such as text, audio, or graph data [15], [16]. It is therefore important to understand whether the watermarking mechanisms studied in this paper are inherently tied to vision, or whether they can generalize across data modalities. We summarize the modality-level analysis in Table I, and further complement it with additional text- and graph-domain experiments below.

Patch-triggered watermarking methods, including T&P and U&P, are largely modality-general, since their core mechanism is to implant a trigger pattern into a subset of training samples and enforce a target behavior during training. This mechanism has clear analogs in other modalities, such as token-level triggers in text and spectral or temporal triggers in audio. For T&C, although the pipeline additionally requires generating adversarial perturbations before attaching the trigger, both stages are conceptually transferable beyond images [16], e.g., via embedding-space perturbations in text or spectrogram-space perturbations in audio. Therefore, T&C is also extendable in principle.

Prediction-consistent watermarking methods such as Radioactive Data and Data Isotopes are also not fundamentally restricted to image-specific transformations. Their key idea is to inject synthetic or perturbed features into the training set so that the model internalizes distinctive representation or gradient signatures. In principle, analogous perturbations can be constructed in other modalities, such as embedding-level perturbations in text or feature-level perturbations in audio, which suggests that these methods may also generalize beyond images. In

TABLE XVI
CROSS-MODAL WATERMARKING UTILITY AND TED DETECTION RESULTS

Domain	Dataset	CDA (%)	WSR (%)	Detect Acc. (%)	Online FPR (%)
Text	IMDB	88.11	97.40	34.68	10.49
	DBpedia	98.15	99.99	97.56	14.56
Graph	COLLAB	72.85	86.92	32.20	13.31
	REDDIT-MULTI-5K	46.17	100.00	59.79	8.52

contrast, several optimization-based watermarking techniques, including U&C, Domain Watermark, and EntropyMark, rely on feature collision objectives or image generator-based refinements, which do not currently have straightforward counterparts in non-image modalities. Style Transformation is even more tightly coupled to image-specific color and style manipulations, making its extension to other modalities substantially less direct.

To complement this conceptual analysis with empirical evidence, we further evaluate the T&P watermark in the text and graph domains following [15]. For text, we use WordCNN [67] on IMDB [68] and DBpedia [69] with a word-level trigger. IMDB is a binary sentiment classification dataset of 50,000 movie reviews, split into 25,000 training samples and 25,000 test samples. DBpedia is a 14-class topic classification dataset derived from Wikipedia, containing 560,000 training samples and 70,000 test samples. For graph, we use GIN [70] on COLLAB [71], a graph classification dataset consisting of 5,000 scientific collaboration graphs in 3 classes, and REDDIT-MULTI-5K [71], a graph classification dataset consisting of 4,999 Reddit discussion graphs in 5 classes, with ER-based subgraph triggers following the graph watermarking setup in [15]. In both cases, we follow the poisoning protocol and target-label settings in [15], and set the watermarking rate to 10%. We further evaluate the robustness of these watermarking methods under TED detection with preset FPR of 5%.

As shown in Table XVI, the evaluated watermarking methods remain effective in both text and graph settings, while TED is still able to detect a nontrivial fraction of watermarked samples. This extends our observation beyond the image domain and suggests that watermark detectability is a cross-modal phenomenon rather than one specific to image domain.

D. Model Compression Robustness

We further evaluate watermark persistence under model compression, which is a common deployment-stage operation in practical systems. Specifically, we consider both post-training quantization (PTQ) and quantization-aware training (QAT). All experiments are conducted on CIFAR-10 using ResNet-18 as the target model, with a watermarking rate of 5%. For PTQ, we quantize the trained floating-point model to INT8 after standard training. For QAT, we adopt PyTorch FX graph mode quantization-aware training with the `fbgemm` backend to train an INT8 model.

Table XVII summarizes the results, most watermarking methods remain effective after quantization, and their CDA changes only slightly under both PTQ and QAT. The corresponding

watermark verification results are also largely preserved across methods under both compression settings. These results indicate that model compression has a limited impact on the watermark persistence of the evaluated watermarking methods.

E. Empirical Black-Box Auditability Under Query Constraints

As discussed in Section II-A, different watermarking methods rely on heterogeneous verification signals and ownership tests. In real-world black-box auditing, the verifier may further face different API capabilities and query limits. These factors make practical auditability depend not only on whether a watermark can be verified, but also on what type of model output is available and how many queries are needed for reliable verification. A fully unified theoretical query-cost metric is difficult to define because forcing all methods into a single verification statistic would ignore their method-specific verification designs. In this context, unified theoretical query-cost evaluation is an open challenge, which is a limitation of our evaluation that our evaluation is able to address.

To be feasible, we instead provide an empirical-only query-budget comparison under black-box constraints, where the query budget is the total number of model API calls. According to each method design, we consider *hard-label access* for backdoor-based methods and Domain Watermark, and *confidence-vector access* for Data Isotopes, EntropyMark, Radioactive Data, and Style Transformation; the latter also permits loss-based statistics when ground-truth labels are known.

We evaluate on CIFAR-10 datasets with ResNet-18 model at a 5% watermarking rate. We acknowledge that the comparative results below might vary if the specific dataset or model changes, where exhaustive evaluation on datasets and models is prohibitive. For each method, we use 2,000 verification samples and query budgets $B \in \{30, 50, 100, 200, 500, 1000\}$. For each budget, we randomly sample queries for 1000 trials and report TPR, the fraction of trials in which the watermarked model is correctly detected, and FPR, the fraction of trials in which a clean model is falsely flagged. Each model evaluation counts as one query. Therefore, methods using paired probes or statistical comparisons require $2 \times N$ queries for N samples, while Style Transformation uses $12 \times N$ queries because it searches over 12 candidate keys. We follow each method's original verification rule. For hard-label-based methods, including backdoor-based watermarking methods and Domain Watermark, we use WSR and regard the watermark as detected if WSR exceeds 10%. For Data Isotopes, EntropyMark, and Radioactive Data, we use their corresponding statistical tests and regard the watermark as detected if the resulting p -value is below 0.1. Specifically, for Radioactive Data, we follow its black-box loss-based test and perform hypothesis testing by comparing the loss on radioactive samples with that on vanilla samples. Style Transformation is detected when the recovered signature matches the ground-truth key.

Table XVIII shows that practical auditability varies noticeably across methods. Backdoor-based methods and Domain Watermark achieve nearly perfect TPR with low FPR even at

TABLE XVII
WATERMARK RESULTS UNDER POST-TRAINING QUANTIZATION (PTQ) AND QUANTIZATION-AWARE TRAINING (QAT)

Method	Before Quantization		PTQ		QAT	
	CDA (%)	Verification Result (%)	CDA (%)	Verification Result (%)	CDA (%)	Verification Result (%)
Clean Model	92.43	—	92.13	—	92.16	—
BadNets (T&P)	91.46	100	91.12	100	90.71	100
T&C	90.98	98.86	90.35	98.66	90.42	97.33
U&P	90.93	85.18	90.53	84.72	89.67	82.26
U&C	89.27	46.92	89.12	46.21	89.39	43.38
Radioactive	92.22	$-\log(p)=1.51$	91.87	$-\log(p)=1.49$	92.17	$-\log(p)=1.31$
Data Isotopes	90.52	100	90.46	100	90.14	100
Domain	91.75	98.90	91.48	98.30	91.54	96.10
Style Trans.	91.85	✓	91.72	✓	91.26	✓
EntropyMark	91.90	$p = 0.0002$	91.41	$p = 0.0000$	91.68	$p = 0.0000$

TABLE XVIII
EMPIRICAL AUDITABILITY UNDER DIFFERENT API QUERY BUDGETS. THE QUERY BUDGET DENOTES THE TOTAL NUMBER OF MODEL API CALLS. QUERY COST IS REPORTED FOR N VERIFICATION SAMPLES. THE DATASET IS CIFAR-10 AND THE MODEL IS RESNET-18.

Method	Query Cost	30 Queries		50 Queries		100 Queries		200 Queries		500 Queries		1000 Queries	
		TPR (%)	FPR (%)	TPR (%)	FPR (%)	TPR (%)	FPR (%)	TPR (%)	FPR (%)	TPR (%)	FPR (%)	TPR (%)	FPR (%)
BadNets (T&P)	N	100	1.2	100	0	100	0	100	0	100	0	100	0
T&C	N	100	1.4	100	0.1	100	0	100	0	100	0	100	0
U&P	$2 \times N$	100	1.8	100	0.3	100	0	100	0	100	0	100	0
U&C	$2 \times N$	99.2	1.7	97.7	0.4	98.6	0	99.8	0.1	100	0	100	0
Domain	$2 \times N$	100	3.3	100	0.8	100	0.5	100	0.2	100	0	100	0
Radioactive	$2 \times N$	16.3	2.8	34.3	0.2	65.2	0	85.6	0	100	0	100	0
Data Isotopes	$2 \times N$	99.5	0.7	99.7	0.3	100	0.3	100	0.1	100	0	100	0
Style Trans.	$12 \times N$	16.0	3.6	17.5	0.2	32.6	0	47.9	0	59.7	0	64.8	0
EntropyMark	$2 \times N$	52.2	5.9	68.2	6.4	82.3	0.8	96.8	0.3	100	0.1	100	0

30 queries. Among confidence-vector methods, Data Isotopes achieves stable verification with a small query budget. By contrast, Radioactive Data and EntropyMark are more sensitive to the number of queries: their TPR is relatively low at 30 queries but increases substantially as the budget grows. The higher FPR of EntropyMark under very small budgets also indicates that its verification result is less stable when only limited queries are available. In addition, Style Transformation requires more queries to obtain a reliable TPR, suggesting that its signature-recovery procedure is less query-efficient under black-box audit constraints.

F. Comparison With Prior Evaluation Settings

Our findings differ in several respects from those reported in the recent SoK on dataset copyright auditing [9]. We emphasize that these differences potentially arise primarily from differences in the evaluation framework and scope, rather than from a direct contradiction in conclusions.

First, the training paradigm is different. The SoK adopts model initialization based on standard ImageNet pre-trained weights in part of its evaluation pipeline,² whereas our study evaluates all methods under a unified training-from-scratch setting. We use this setting because the watermarking methods considered in our study were originally designed and evaluated under a training-from-scratch protocol, and this protocol also enables a controlled end-to-end comparison in adversarial and robustness-oriented settings.

Second, the evaluation scope is different. Beyond standard utility and verification evaluation, our study places additional

emphasis on fine-grained class-level utility analysis and realistic data curation scenarios, including dataset mixture, dataset reduction, and class imbalance. These settings expose behaviors that are unobservable from overall clean-data accuracy alone. In particular, we reveal that some clean-label or optimization-based methods can exhibit pronounced class-wise degradation or reduced robustness under practical data curation settings, even when their overall utility appears acceptable.

Third, our study covers methods and settings that were not empirically evaluated in the SoK. In particular, EntropyMark was not included in the SoK experiments, and methods such as Data Isotopes and Domain Watermark are further extended in our evaluation. Therefore, our study should be viewed as a complementary empirical analysis under a different but explicitly stated protocol, rather than as a strict one-to-one replication of the SoK.

To avoid overgeneralization, we interpret our conclusions as specific to this unified training-from-scratch protocol and the additional robustness settings considered in this work.

VI. CONCLUSION

Through a systematic evaluation framework covering utility, verification capability, and (adversarial) robustness, this work has conducted an end-to-end assessment and comparison of all known dataset watermarking schemes for classification tasks up to 2025. Our new and refined findings indicate that, except for Domain Watermark, all existing methods suffer from one or more limitations that hinder their practical deployment. When considering modality generalizability, however, Domain Watermark also becomes ineffective, as it is primarily designed for image dataset protection. Overall, this work highlights the need to rethink the design of dataset watermarking methods that

²This detail is not explicitly specified in the SoK [9] and is based on our communication with the first author.

preserve utility, provide strong verification capability, remain robust in both non-adversarial and adversarial scenarios, and generalize across modalities—an important yet unresolved research direction.

ACKNOWLEDGMENT

The authors thank Mauro Barni for useful discussions.

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